

PRACTICAL COURSE WORKBOOK

Python Fundamentals

Write, test and explain a small Python program

LEVEL	Beginner
GUIDED TIME	5 hours across 12 lessons
PROJECT	a stock-report script with fictional data, tests and run instructions
SKILLS	Python, OOP, File I/O, Critical evaluation, Documentation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Run a Python file and interpret its output.
- Use conditions, loops, collections and functions to process records.
- Validate input and read CSV data without hiding errors.
- Write and run tests for normal and failure cases.
- Build a small stock-report program with a clear handover.

WHO THIS IS FOR

Beginners who want to solve small tasks with code, with no previous programming experience required.

BEFORE YOU BEGIN

- Create and locate a plain-text file
- Use a terminal or a browser-based Python 3 environment

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

Write, run and debug small Python programs. Learn values, decisions, loops, collections, functions, file handling, classes and tests, then build a stock report from fictional CSV data.

MODULE	LESSONS AND FOCUS
01 First programs	1. Run a Python file and inspect the result 2. Separate text from numeric values 3. Make a decision with if and elif
02 Collections and reusable code	1. Process a collection with a loop 2. Organise records with lists and dictionaries 3. Write a function that returns a useful value
03 Reliable input and files	1. Validate a whole-number input in Python 2. Read CSV records without splitting them by hand 3. Load JSON and check its structure
04 Build and test a small application	1. Keep instance data separate in a small class 2. Test a rule at its boundary 3. Build a stock report from validated records

HOW TO USE EACH LESSON PAGE

- 1 Read the objective and identify the decision it supports.
- 2 Attempt the retrieval prompt before checking the online explanation.
- 3 Reproduce the worked example with the stated input.
- 4 Test one normal case and one boundary or failure case.
- 5 Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

First programs

Run a Python file and inspect the result; Separate text from numeric values; Make a decision with if and elif.

LESSON 01 / 18 MIN

Run a Python file and inspect the result**Learning objectives**

- Save and run a Python file.
- Distinguish source code from output.
- Check a calculation independently.

Key terms

File, Interpreter, Output

Retrieval prompt

You changed the code but see the old output. What should you check first?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

LESSON 02 / 22 MIN

Separate text from numeric values**Learning objectives**

- Identify text and numeric values.
- Convert valid whole-number text.
- Explain a formatted output.

Key terms

String, Integer, Formatting

Retrieval prompt

What does '12' + '3' produce?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

LESSON 03 / 26 MIN

Make a decision with if and elif**Learning objectives**

- Write an ordered decision chain.
- Use a comparison at a threshold.
- Test each branch and the equality case.

Key terms

Condition, Branch, Boundary

Retrieval prompt

With quantity = 5 and reorder_level = 5, which comparison includes the boundary?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

Collections and reusable code

Process a collection with a loop; Organise records with lists and dictionaries; Write a function that returns a useful value.

LESSON 04 / 30 MIN

Process a collection with a loop

Learning objectives

- Visit each list item once.
- Maintain a running total.
- Explain the empty-list result.

Key terms

Iterable, Loop, Accumulator

Retrieval prompt

What is total when quantities is [] in this example?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

LESSON 05 / 18 MIN

Organise records with lists and dictionaries

Learning objectives

- Represent several records in a list.
- Look up fields by dictionary key.
- Distinguish a required field from an optional one.

Key terms

List, Dictionary, Record

Retrieval prompt

What happens when a dictionary literal repeats the same key?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

LESSON 06 / 22 MIN

Write a function that returns a useful value

Learning objectives

- Pass values into a function.
- Return a Boolean result.
- Separate calculation from presentation.

Key terms

Argument, Function, Return value

Retrieval prompt

A function reaches its end without executing return. What does its caller receive?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

Reliable input and files

Validate a whole-number input in Python; Read CSV records without splitting them by hand; Load JSON and check its structure.

LESSON 07 / 26 MIN

Validate a whole-number input in Python

Learning objectives

- Convert text to an integer and handle ValueError.
- Reject zero and negative quantities explicitly.
- Test valid, invalid and boundary inputs.

Key terms

validation, Python, data

Retrieval prompt

Why does '-2' need a check after int() succeeds?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

LESSON 08 / 30 MIN

Read CSV records without splitting them by hand

Learning objectives

- Read quoted CSV fields correctly.
- Use headers to identify columns.
- Convert numeric text before arithmetic.

Key terms

CSV, Header, Conversion

Retrieval prompt

Why should you avoid line.split(',') for this fixture?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

LESSON 09 / 18 MIN

Load JSON and check its structure

Learning objectives

- Parse a JSON document.
- Check its container and field types.
- Reject an invalid settings value.

Key terms

JSON, Shape, Validation

Retrieval prompt

What does successful json.loads() establish?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

Build and test a small application

Keep instance data separate in a small class; Test a rule at its boundary; Build a stock report from validated records.

LESSON 10 / 22 MIN

Keep instance data separate in a small class

Learning objectives

- Create two independent objects.
- Store instance attributes in `__init__`.
- Call a method using an object's data.

Key terms

Class, Instance, Method

Retrieval prompt

Which change should update only the books instance?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

LESSON 11 / 26 MIN

Test a rule at its boundary

Learning objectives

- Encode an expected result in a test.
- Cover an equality boundary.
- Recognise a regression after changing a rule.

Key terms

Requirement, Test case, Regression

Retrieval prompt

Which input exposes changing `<=` to `<` in the reorder rule?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

LESSON 12 / 30 MIN

Build a stock report from validated records

Learning objectives

- Validate records before reporting.
- Produce a deterministic list of selected items.
- Document how to run and check a small program.

Key terms

Input contract, Selection, Handover

Retrieval prompt

An invalid row is found while building a stock report. What is the clearest policy for this example?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Build a stock report from a fictional CSV file.

DELIVERABLE	stock_report.py, a sample CSV, tests and a short README with example output.
TOOLS	Python 3, Text editor, Terminal, Fictional CSV file

Production plan

- 1 Define CSV columns for item name, quantity and reorder level.
- 2 Parse quantities as integers and reject missing names or negative values with a useful message.
- 3 Write a function that finds items at or below their reorder level.
- 4 Test empty data, a boundary quantity, invalid numeric text and a normal record.
- 5 Print a readable report and document how to run the script and tests.

Definition of done

- The same input produces the same ordered report.
- Boundary and invalid inputs have documented behaviour.
- Empty files and bad headers do not produce a misleading success message.
- Tests run with one documented command.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER

State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS

List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE

Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can someone run this from my instructions?
- Are malformed records reported clearly?
- Do the tests check a real failure and a boundary?

FINAL DECISION

Is the project ready to share? State the evidence, remaining risk, owner and next review date.

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Ask another learner to run your program with one new CSV fixture, then improve the first confusing error message.

PRIMARY REFERENCES

Sources to revisit

An informal introduction to Python

Python Software Foundation - reviewed 2026-10-07

<https://docs.python.org/3/tutorial/introduction.html>

Python language reference

Python Software Foundation - reviewed 2026-10-07

<https://docs.python.org/3/reference/index.html>

Built-in functions: int and print

Python Software Foundation - reviewed 2026-10-07

<https://docs.python.org/3/library/functions.html>

Control flow: if statements

Python Software Foundation - reviewed 2026-10-07

<https://docs.python.org/3/tutorial/controlflow.html>

Built-in types: sequence operations

Python Software Foundation - reviewed 2026-10-07

<https://docs.python.org/3/library/stdtypes.html>

Data structures: dictionaries

Python Software Foundation - reviewed 2026-10-07

<https://docs.python.org/3/tutorial/datastructures.html>

Defining functions

Python Software Foundation - reviewed 2026-10-07

<https://docs.python.org/3/tutorial/controlflow.html#defining-functions>

Errors and exceptions

Python Software Foundation - reviewed 2026-10-07

<https://docs.python.org/3/tutorial/errors.html>

csv: CSV file reading and writing

Python Software Foundation - reviewed 2026-10-07

<https://docs.python.org/3/library/csv.html>

json: JSON encoder and decoder

Python Software Foundation - reviewed 2026-10-07

<https://docs.python.org/3/library/json.html>

Classes and instance variables

Python Software Foundation - reviewed 2026-10-07

<https://docs.python.org/3/tutorial/classes.html>

unittest: unit testing framework

Python Software Foundation - reviewed 2026-10-07

<https://docs.python.org/3/library/unittest.html>

Input and output

Python Software Foundation - reviewed 2026-10-07

<https://docs.python.org/3/tutorial/inputoutput.html>